



BSI Standards Publication

Cranes — Competency requirements for crane inspectors

National foreword

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A list of organizations represented on this committee can be obtained on request to its committee manager.

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International Standard

ISO 23814

Cranes — Competency requirements for crane inspectors

*Appareils de levage à charge suspendue — Exigences relatives
aux compétences pour les inspecteurs d'appareils de levage à
charge suspendue*

**Second edition
2024-07**



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Foreword

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The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO document should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

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This document was prepared by Technical Committee ISO/TC 96, *Cranes*, Subcommittee SC 5, *Use, operation and maintenance*.

This second edition cancels and replaces the first edition (ISO 23814:2009), which has been technically revised.

The main changes are as follows:

- in [4.2](#), technical knowledge to be possessed by crane inspectors has been revised;
- Clause 5 has been removed.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Introduction

The objective of this document is to achieve uniform competency standards for work involving crane inspectors.

In some countries, additional requirements for the competency of crane inspectors can apply.

Cranes — Competency requirements for crane inspectors

1 Scope

This document specifies the competency required for persons who carry out periodic, exceptional, and enhanced periodic inspections of cranes. It excludes the inspection and checks performed by crane operators and maintenance personnel.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 4306-1, *Cranes — Vocabulary — Part 1: General*

ISO 4310, *Cranes — Test code and procedures*

ISO 9927 (all parts), *Cranes — Inspections*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 4306-1, ISO 4310 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <https://www.electropedia.org/>

3.1

crane inspector

person with the necessary knowledge, training and experience to perform the specific crane inspection

Note 1 to entry: [Annex A](#) provides categories of crane inspectors.

4 Requirements

4.1 Independence, impartiality and integrity

Inspectors shall avoid undue influences or conflicts of interest that can compromise professional independence, ethics, objectivity or inspection integrity.

4.2 Technical knowledge

Crane inspectors shall possess the technical knowledge of the following items and shall be proficient at inspection methods as applicable to the assigned tasks:

- a) structures and components:
 - 1) structural members and connecting elements;

- 2) electrical elements including power connections, motors, circuits, control systems, sensors and electronics, etc.;
 - 3) mechanical components including gearboxes, drums, bearings, brakes, clutches, sheaves, ropes/chains with their fittings, lifting attachments, etc.;
 - 4) hydraulic components including hoses, cylinders, valves and fittings, etc.;
 - 5) internal combustion engines;
 - 6) lubrication systems;
- b) assembly and operation:
- 1) proper assembly including installation of connecting elements, etc.;
 - 2) reeving and drum storage;
 - 3) safe access including ladders, hatches, platforms and guardrails, etc.;
 - 4) cabs and control stations;
 - 5) safety labels;
 - 6) basic crane functions and operations;
- c) documentation:
- 1) maintenance schedules and procedures;
 - 2) record keeping;
 - 3) crane manuals and service bulletins;
 - 4) specifications and certifications for replaceable hardware such as ropes and bolts;
- d) safety, rules and practices:
- 1) applicable international standards (e.g. ISO 12480-1);
 - 2) site-specific and equipment-specific safety rules;
 - 3) use of personal protective equipment;
- e) tools and methods:
- 1) use, care, calibration and maintenance of inspection tools, including sheave gages, dial gage, calliper, torque wrench and tensioning devices, etc.;
 - 2) non-destructive testing;
 - 3) load tests, calibration and limits.

Crane inspectors should possess the technical knowledge of national safety regulations related to cranes.

Performance criteria are given in [Annex B](#).

4.3 Experience requirements

The crane inspector shall possess knowledge and skills relevant to the equipment to be inspected and the methods employed. Experience can be gained through a combination of training, apprenticeship, and collateral performance of crane design, manufacture, installation, maintenance or operation.

4.4 Techniques for crane inspection

Crane inspectors shall be competent in inspecting cranes and crane components in accordance with the ISO 9927 series, as applicable to the type of crane.

Annex A (informative)

Recommended crane inspector categories

The following categories are recommended:

- a) tyre-mounted mobile crane inspector;
- b) crawler-mounted mobile crane inspector;
- c) tower crane inspector;
- d) jib crane inspector;
- e) bridge and gantry crane inspector;
- f) inspector for other specific types of crane for which the inspector has been given training.

Annex B

(informative)

Performance criteria for technical knowledge

See [Table B.1](#).

Table B.1 — Performance criteria related to technical knowledge

Item (see 4.2)	Performance criteria
a) Structures and components	Knowledge of 1) structural members and connecting elements — corrosion limits — alignment and distortion tolerances — welds — bolting material handling, specifications and discard criteria 2) electrical elements including power connections, motors, circuits, control systems, sensors and electronics, etc. — electrical wiring diagrams — maintenance and protection of wiring, terminals, electrical enclosures and devices — protection and support of power cables — grounding 3) mechanical components including gearboxes, drums, bearings, brakes, clutches, sheaves, ropes/chains with their fittings, lifting attachments, etc. — fit and tolerances — alignment — maintenance and lubrication requirements — brake and clutch wear limits — sheave groove profile and wear limits — rope/chains construction, dimensional tolerances, defects and discard criteria, proper installation, handling and maintenance — lifting attachments inspection and discard criteria

Table B.1 (continued)

Item (see 4.2)	Performance criteria
	4) hydraulic components including hoses, cylinders, valves and fittings, etc. — circuits, fluid flow and valve functionality — types of pumps and their control systems — system performance and leakage — hose specification — hydraulic fluid specifications — filtration and cleanliness — cooling 5) internal combustion engines — service and maintenance — mounting and vibration — defects and causes 6) lubrication systems — types of systems — lubricants — performance assessment
b) Assembly and operation	Knowledge of 1) proper assembly including installation of connecting elements, etc. — assembly diagrams — bolt installation and tensioning — pin installation and tolerances — plumbness 2) reeving and drum storage — reeving diagrams — spooling fundamentals 3) safe access including ladders, hatches, platforms and guardrails, etc.

ISO 23814:2024(en)**Table B.1** *(continued)*

Item (see 4.2)	Performance criteria
	4) cabs and control stations — window requirements — guards — controls and instruments 5) safety labels 6) basic crane functions and operations — reading and applying the load capacity chart — motions — motive power and transmission — control functions — limits — winches — housekeeping
c) Documentation	Knowledge of 1) maintenance schedules and procedures 2) record keeping 3) crane manuals and service bulletins 4) specifications and certifications for replaceable hardware such as ropes and bolts
d) Safety, rules and practices	Knowledge of 1) applicable international standards (e.g. ISO 12480-1) 2) site-specific and equipment-specific safety rules 3) use of personal protective equipment — fall protection fundamentals — fall arrest systems

Table B.1 (continued)

Item (see 4.2)	Performance criteria
e) Tools and methods	Knowledge of 1) use, care, calibration and maintenance of inspection tools, including sheave gages, dial gage, calliper, torque wrench and tensioning devices, etc. 2) non-destructive testing, working knowledge of methods and their uses, and interpretation of test results — visual inspection — ultrasonic testing — magnetic particle inspection — dye penetrant — other methods e.g. X-ray, eddy current 3) load tests, calibration and limits — test weight certification — load test methods and acceptance criteria — ranges for limits — calibration tolerances

Bibliography

- [1] ISO 12480-1, *Cranes — Safe use — Part 1: General*

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